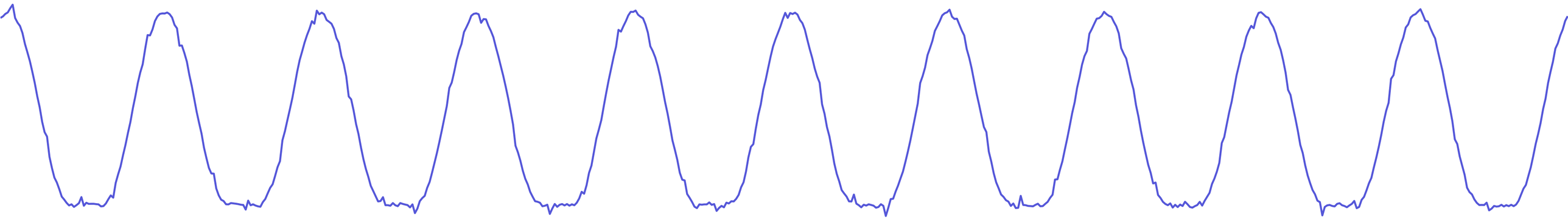
UNIVERSITÀ DEGLI STUDI DI MILANO

Biomedical Signal Processing · 2025 / 2026

Heart rate and respiratory rate

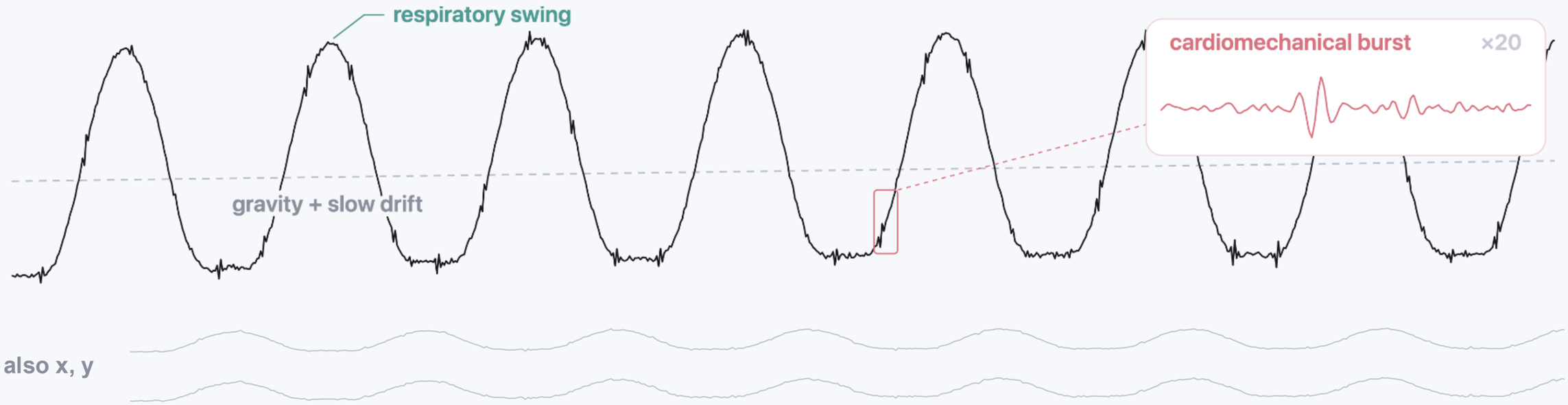
from a single chest accelerometer

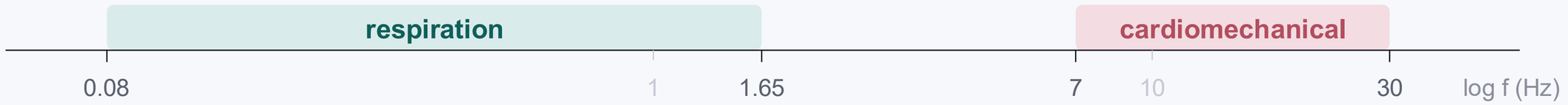


Enrico Dalla Stella – 84266A

THE PROBLEM

What the accelerometer records





OVERVIEW

From acceleration to physiological information

Chest acceleration

x, y, z



01

Sampling analysis

median T_s from timestamps, resample on jitter

02

Trim · detrend

5 s off each end, linear trend removed

03

Band separation

4th-order zero-phase Butterworth

Respiration
periodicity

PC1 (PCA)



→ **Autocorrelation → period** → **Welch + persistence** **RR**

Heart
events

Nonlinear envelope



→ **Template → NCC** → **Beat times → IBI** **HR**

Reliability gate supported by an independent estimate and consistency checks — otherwise the value is withheld

METHOD

Preprocessing and band separation

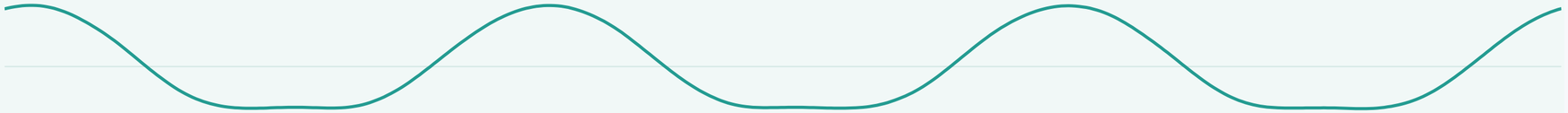
01 **Raw axis**
gravity, drift, transients



02 **Detrended**
zero mean, trend removed



03 **Respiratory band**
0.08 – 1.65 Hz



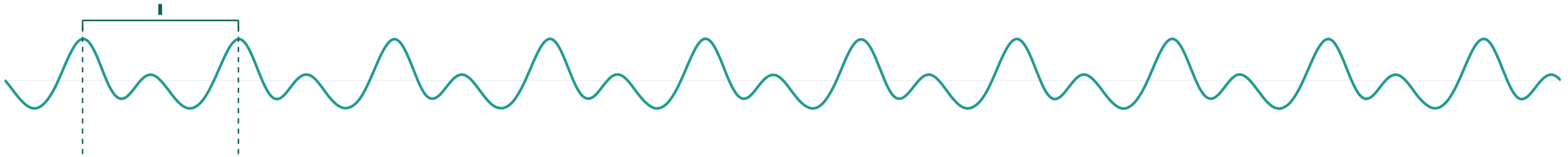
04 **Cardiac band**
7 – 30 Hz



Filtering selects a time scale. It does not identify the source.

Respiratory rate: periodicity

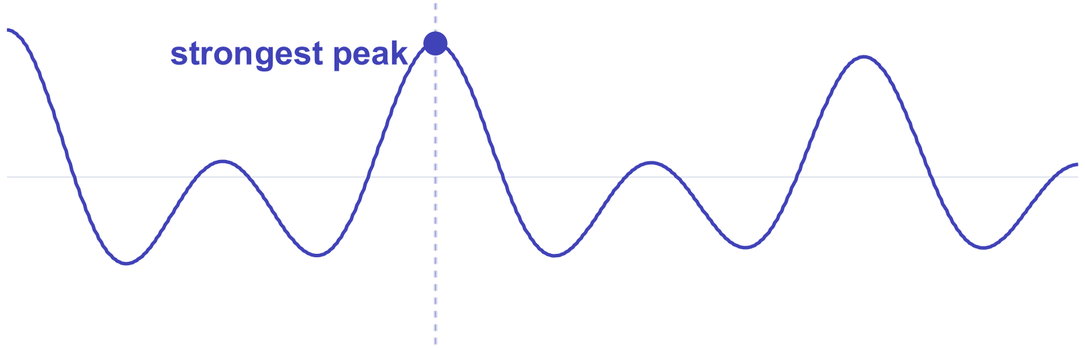
PC1 one series



Autocorrelation

primary estimate

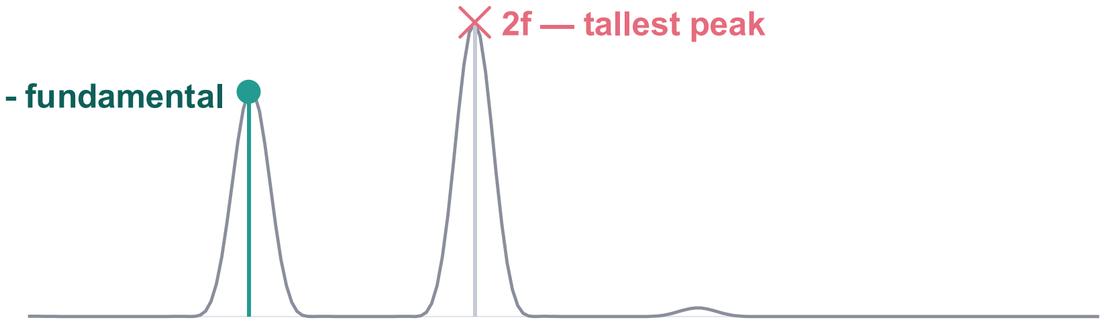
After how long does the signal resemble itself?



Welch PSD

consistency check

Which frequencies carry repeated energy?



T = 4.0 s → f = 1/T = 0.25 Hz → RR = 60/T

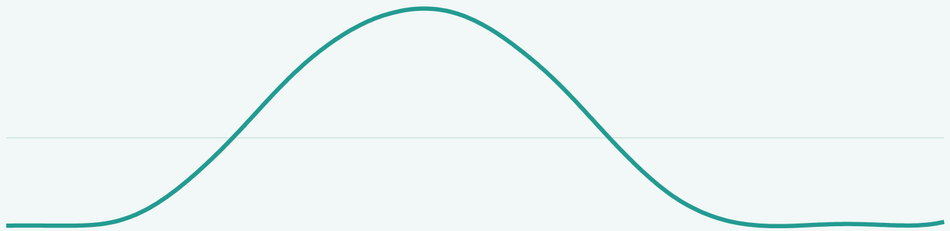
15.0 breaths / min

TRANSITION

Why the heart is different

Respiration

one smooth oscillation



The period **is** the physiological variable

Periodicity estimated directly on the band

Heart

a train of short bursts



Periodicity is **supporting evidence**

Individual events still have to be localised

Same time span, same recording — two phenomena that do not tolerate the same estimator.

METHOD · HEART

From energy bursts to detected beats

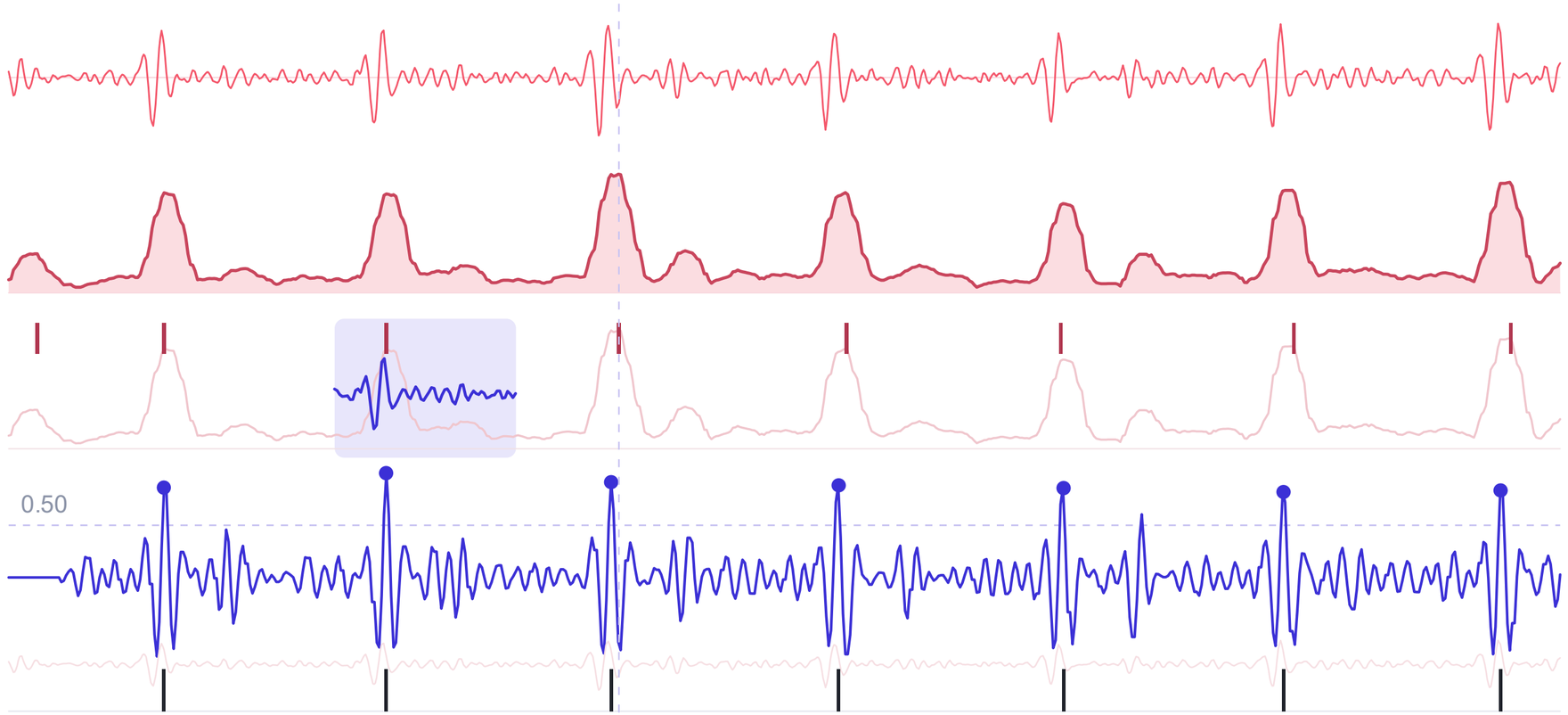
01 **Cardiac band**
7 – 30 Hz

02 **Nonlinear envelope**
fourth power · 3 Hz LP

03 **Candidates → template**
−0.20 s … +0.50 s

04 **NCC with template**
shape, not amplitude

05 **Beat times → IBI**
accepted maxima



METHOD · HEART

From beats to heart rate

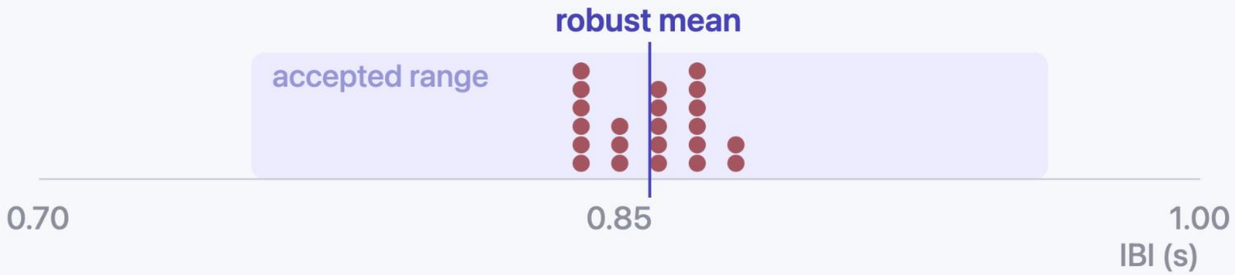
Detected beats → intervals

primary path



Interval distribution

robust mean, outliers dropped



HR = 60 / mean(IBI) **70** bpm

Published only with support

independent check

PERIODICITY

RMS-envelope ACF supports the IBI-derived rate

SPACE

Cross-axis agreement strengthens the periodicity evidence

TIME

Local persistence supports the ACF fallback

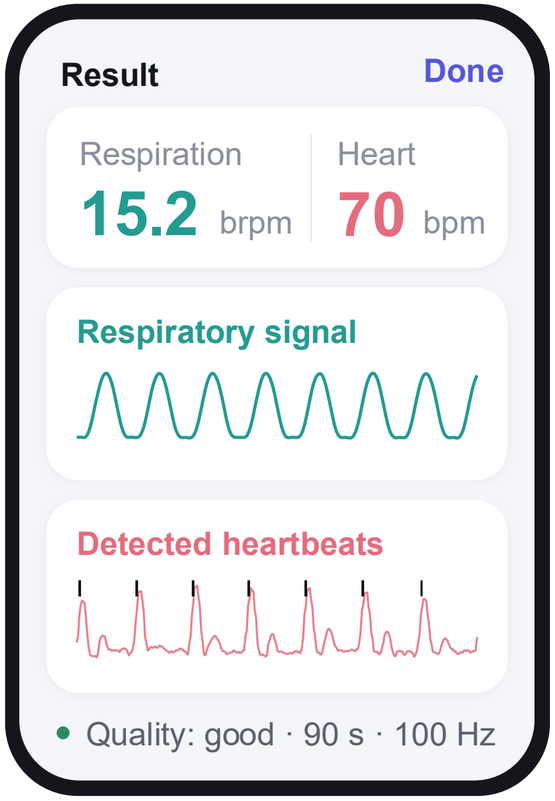
> Stable cross-axis ACF can provide a window-averaged HR fallback when beat localisation is unreliable.

Otherwise: no value.

An unresolved f / 2f case ends in abstention, not in a multiply-by-two rule.

IMPLEMENTATION

The pipeline in a real system



Core Motion → DSP pipeline → RR / HR + quality → Pulse UI

- Guided 60 / 90 / 120 s measurement
- Swift DSP mirrors the offline Python harness
- Raw per-session CSV export for re-analysis
- No account, no network, no analytics

CLOSE

Conclusions

The contribution is the decision strategy, not one isolated DSP block.

ONE SENSOR, TWO ESTIMATION PROBLEMS

Respiration

Periodicity is the quantity of interest

Estimate the correct cycle despite harmonics

Heart

Periodicity supports HR

Localise events, then derive HR from timing

THE DESIGN PRINCIPLE

No single feature is trusted alone.

- time-domain periodicity
- spectral consistency
- morphology
- agreement across axes
- persistence over time

TAKE-HOME MESSAGE

The challenge was not finding periodicity in the accelerometer signal, but deciding when that periodicity could legitimately be interpreted as physiology.

Next step: quantify physiological accuracy through systematic reference-based validation on human recordings.

Thank you.